

Parity families and a kernel-averaged L -function for near-Ramanujan signings

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July 2026

Abstract

For a signing σ of a d -regular graph, the spectrum of A_σ depends only on the signs of cycles. We study the affine $\mathbb{F}_2$ family of signings making every short even cycle unbalanced, and show that averaging over it converts the sign problem of the Bilu-Linial conjecture into a counting problem: a master identity expresses the family-averaged trace as a parity-weighted sum over wrap classes confined to the span W of the constraint cycles, and the family-averaged Ihara L -function diagonalizes so that every prime whose parity escapes W contributes the Ramanujan rate $\sqrt{d-1}$ automatically. Uniform averaging over all signings, by contrast, provably cannot certify a spectral radius below the Kesten profile. We prove matched upper and lower bounds for the confined walk counts, a doubling injection from below, and from above an ear-decomposition encoding in which the number of fresh runs of a non-backtracking walk equals the cycle rank of its support, combined with a window lemma for bicycle-free graphs and a rank bound via the Moore bound for irregular graphs. Consequences include ε -versions of the Bilu-Linial conjecture: every d -regular graph that is subcritical at scale $\log n$, and every d -regular graph bicycle-free at radius $C \log \log n / \delta$, admits a signing in the parity family with $\rho(A_\sigma) \leq 2\sqrt{d-1}(1 + C\delta \log(1/\delta))(1 + o(1))$. We further identify the necessary hypotheses exactly (K_d -trapping; tree-burst gadgets), give an exact certificate on the hypercube, and record a decisive obstruction to two-sided interlacing: $\mathbb{E}_\sigma \det(xI - A_\sigma^2)$ is not real-rooted, already for the quadrilateral, where it equals $(x^2 - 4x + 2)^2 + 4$.

1 Introduction

Bilu and Linial [3] conjectured that every d -regular graph admits a signing $\sigma: E \rightarrow \{\pm 1\}$ of its adjacency matrix with $\rho(A_\sigma) \leq 2\sqrt{d-1}$, the spectral radius of the infinite d -regular tree. Since the new eigenvalues of a 2-lift are the eigenvalues of a signed adjacency matrix, the conjecture would produce infinite towers of d -regular Ramanujan graphs of every degree. Marcus, Spielman and Srivastava [11] proved the one-sided bound $\lambda_{\max}(A_\sigma) \leq 2\sqrt{d-1}$ via interlacing families, resolving the bipartite case; Mohanty, O'Donnell and Paredes [12] showed that a uniformly random signing is near-Ramanujan when every ball of radius $r \gg (\log \log n)^2$ contains at most one cycle; and Xu and Zhang [13] recently proved the two-sided bound $2\sqrt{3(d-1)}$ for all graphs of maximum degree d . Even the ε -relaxed conjecture remains open in general, and the obstruction is concentrated on graphs that are locally rich in short cycles.

*Verification suite, drivers, and experimental data: <https://github.com/Vaibhavs25/bilu-linial-parity>

This paper develops a method adapted to exactly that regime. The spectrum of A_σ depends only on the switching class of σ , i.e. on the signs of cycles, and short *even* cycles enter the even traces, hence the spectral radius, at first order, while odd cycles enter only in pairs (Lemma 1). We therefore work with the affine $\mathbb{F}_2$ solution family $\mathcal{F}$ of the system “every even cycle of length $\leq L$ is unbalanced”. Averaging over $\mathcal{F}$ has two structural consequences. First, a *master identity* (Proposition 7): the family-averaged trace equals a parity-weighted sum of walk counts over wrap classes confined to the span W of the constraint cycles: every walk whose parity touches a long cycle is annihilated, single constraint cycles enter with a deterministic -1 , pairs with $+1$. Second, a *kernel-averaged L -function* (Proposition 10): Bass’s Euler product diagonalizes under the family average, the primes whose parity escapes W contributing exactly $\frac{1}{2} \log \det(I - u^2 B)$, which is analytic precisely in the Ramanujan disk $|u| < (d - 1)^{-1/2}$. The entire deviation of the averaged L -function from the Ramanujan rate is carried by parity-confined primes: at the level of this identity the sign problem becomes a counting problem (Question 11), while the quantitative proofs below run through the trace certificate of Corollary 8 at $k \asymp \log n$. By contrast, the uniform average over all signings retains only the totally even walks, whose number always dominates the tree profile (Corollary 8): uniform averaging can never certify a sub-Kesten signing, while the family average can and, in computation, does.

Our main theorems bound the confined counts and convert them into signings. From below, a doubling injection shows the totally even non-backtracking walks grow at rate exactly $\sqrt{d-1}$ (Proposition 12); from above we prove a matching bound whose engine is the observation that in a non-backtracking walk the maximal runs of fresh edges are precisely the ears of an ear decomposition of the support, so their number equals its cycle rank (Proposition 16). Two further ingredients extend the count from bounded rank to bounded cycle density: a *window lemma*: in a graph whose R -balls are at most unicyclic, a non-backtracking path is determined per R -window by endpoints, direction and winding, so paths grow at rate $\exp(O(\log(sR)/R))$ (Lemma 19), and a *rank lemma* via the Moore bound for irregular graphs [1]: bicycle-free graphs with s edges have cycle rank at most $1 + Cs \log s/R$ (Lemma 20), the $\log s$ being necessary because high-girth cages are bicycle-free with rank density approaching one. The resulting ε -theorems (Propositions 24 and 21) produce, for every d -regular graph that is subcritical at scale $\log n$ or bicycle-free at radius $C \log \log n/\delta$, a signing in the parity family with $\rho(A_\sigma) \leq 2\sqrt{d-1}(1 + C\delta \log(1/\delta))(1 + o(1))$; the hypothesis scale of [12], reached by an independent route, with the conclusion holding inside the constrained family (including $W = \{0\}$, which recovers a near-Ramanujan statement for random signings).

The hypotheses are shown to be necessary in a strong sense: a K_d subgraph traps totally even walks at rate $d-2 > \sqrt{d-1}$ (Proposition 13), and tree-burst gadgets realize the window lemma’s rate, so both the core exclusion and the $\log \log n$ radius are sharp in form. Along the way we give an exact certificate on the hypercube: every solution of the quadrilateral system satisfies $A_\sigma^2 = nI$, recovering Huang’s signing [9] as a special case of the family (Proposition 5), and we record why the natural two-sided extension of the interlacing method fails at the first gate: $\mathbb{E}_\sigma \det(xI - A_\sigma^2)$ is not real-rooted, with the exact identity $q_{C_4} = (x^2 - 4x + 2)^2 + 4$ (Remark 6). All identities and bounds in this paper were additionally verified by exact computation; the experiments, which guided the constructions and falsified two intermediate conjectures, are summarized in Section 7. Exact spectral results for signed circulants arising from the same family appear in a companion note [4].

2 Setup

$G = (V, E)$ is a simple connected graph, $|V| = n$, $|E| = m$, adjacency matrix A . A signing is $\sigma \in \{\pm 1\}^E$, encoded by the bit vector $s \in \mathbb{F}_2^E$ with $\sigma_e = (-1)^{s_e}$; the signed adjacency matrix A_σ has entries $\sigma_{uv} A_{uv}$. We write $\rho(A_\sigma) = \max_i |\lambda_i(A_\sigma)|$. The cycle space $Z(G) \leq \mathbb{F}_2^E$ has dimension $m - n + 1$; the *sign* of $z \in Z(G)$ under σ is $(-1)^{s \cdot z}$, and a cycle C is *unbalanced* if $s \cdot \chi_C = 1$. Switching preserves the spectrum and acts trivially on cycle signs; conversely the switching class of σ is exactly the linear functional $z \mapsto s \cdot z$ on $Z(G)$, so there are 2^{m-n+1} switching classes.

For d -regular G the conjecture of Bilu and Linial [3] asserts $\min_\sigma \rho(A_\sigma) \leq 2\sqrt{d-1}$, the spectral radius of the infinite d -regular tree. Marcus, Spielman and Srivastava [11] proved the one-sided bound $\lambda_{\max}(A_\sigma) \leq 2\sqrt{d-1}$ for some σ , resolving the bipartite case; Mohanty, O'Donnell and Paredes [12] showed a random signing attains $\rho \leq 2\sqrt{d-1} + \varepsilon$ when every ball of radius $r \gg (\log \log n)^2$ contains at most one cycle; and Xu and Zhang [13] recently proved $\min_\sigma \rho(A_\sigma) \leq 2\sqrt{3(d-1)}$ for all graphs of maximum degree d via interlacing families of mixed characteristic polynomials.

3 Even-trace parity

For a closed walk W of length ℓ let $m_e(W)$ be the number of traversals of e , and let $z(W) \in \mathbb{F}_2^E$ be the parity vector $z(W)_e = m_e(W) \bmod 2$. Since W is closed, every vertex has even total incidence with the edge multiset of W , so the support of $z(W)$ has all even degrees: $z(W) \in Z(G)$. Moreover $|z(W)| \equiv \sum_e m_e(W) = \ell \pmod{2}$, and the sign of W under σ is $\prod_e \sigma_e^{m_e(W)} = (-1)^{s \cdot z(W)}$.

Let $Z_{\text{ev}} = \{z \in Z(G) : |z| \text{ even}\}$, the kernel of the linear map $z \mapsto |z| \bmod 2$ on $Z(G)$. If G is bipartite then $Z_{\text{ev}} = Z(G)$; otherwise Z_{ev} has index 2, hence dimension $m - n$.

Lemma 1 (Even-trace parity). *For every signing σ and every even ℓ ,*

$$\text{tr}(A_\sigma^\ell) = \sum_{z \in Z_{\text{ev}}} N_\ell(z) (-1)^{s \cdot z},$$

where $N_\ell(z) \geq 0$ is the number of closed ℓ -walks in G with parity vector z , independent of σ .

Proof. Group the closed ℓ -walks by $z(W)$ and use $\text{sign}(W) = (-1)^{s \cdot z(W)}$. For even ℓ only classes with $|z|$ even occur, by the congruence above. $\square$

Corollary 2. $\rho(A_\sigma)$ is a function of the restriction of the character $z \mapsto (-1)^{s \cdot z}$ to Z_{ev} . In particular $\rho(A_{-\sigma}) = \rho(A_\sigma)$, and for non-bipartite G the map from switching classes to ρ -relevant data is 2-to-1: at most 2^{m-n} values of ρ occur among the 2^{m-n+1} classes.

Proof. $\rho(A_\sigma) = \lim_{k \rightarrow \infty} \text{tr}(A_\sigma^{2k})^{1/2k}$ since $\rho^{2k} \leq \text{tr}(A_\sigma^{2k}) \leq n\rho^{2k}$, and by Lemma 1 every even trace depends on s only through $s|_{Z_{\text{ev}}}$. Negating σ adds $|z| \bmod 2$ to $s \cdot z$, which vanishes on Z_{ev} . $\square$

Corollary 3 (Interference accounting). *Let C be a cycle of G . If $|C|$ is even, the class $z = \chi_C$ contributes to even traces with the sign $(-1)^{s \cdot \chi_C}$ of C itself (first order). If $|C|$ is odd, then $\chi_C \notin Z_{\text{ev}}$ and C enters even traces only through combined classes; the lowest-order such classes are $z = \chi_{C_1} + \chi_{C_2}$ for pairs of odd cycles, contributing with sign equal to the product of the two cycle signs.*

Corollary 3 identifies the sign structure but not the magnitudes $N_\ell(z)$, so it does not by itself prove monotonicity statements. It does, however, explain two robust empirical phenomena (§7): making all short *even* cycles unbalanced places a deterministic (-1) on their first-order terms, while making all short *odd* cycles unbalanced makes every odd-pair term $(+1)(+1)$ or $(-1)(-1)$, i.e. sign-definite *positive*, eliminating cancellation; and indeed exhaustive optima place roughly half of all triangles unbalanced, while forcing all triangles unbalanced raises ρ above the random-signing mean.

4 An $\mathbb{F}_2$ obstruction and the parameter β_L

Fix L and consider the linear system over $\mathbb{F}_2$

$$s \cdot \chi_C = 1 \quad \text{for every even cycle } C \text{ of length } \leq L. \quad (1)$$

Lemma 4 (K_4 obstruction). *If G contains K_4 as a subgraph, then the system (1) with $L \geq 4$ is inconsistent: no signing of G makes all quadrilaterals unbalanced.*

Proof. Let $\{a, b, c, d\}$ span a K_4 and let C_1, C_2, C_3 be its three 4-cycles. Every edge of the K_4 lies in exactly two of them, so $\chi_{C_1} + \chi_{C_2} + \chi_{C_3} = 0$ in $\mathbb{F}_2^E$, whence $s \cdot \chi_{C_1} + s \cdot \chi_{C_2} + s \cdot \chi_{C_3} = 0$ for every s , while (1) demands the sum be $1 + 1 + 1 = 1$. $\square$

More generally, any odd-cardinality dependent family of even short cycles blocks (1). Define

$$\beta_L(G) = \frac{1}{\#\{\text{even cycles } \leq L\}} \max_{s \in \mathbb{F}_2^E} \#\{\text{even cycles } C, |C| \leq L : s \cdot \chi_C = 1\},$$

the maximum simultaneously-unbalanceable fraction. Every explicit signing certifies a lower bound on β_L , and $\beta_L \geq \frac{1}{2}$ always, by averaging over uniform signings; row reduction produces the affine solution family of a consistent subsystem, whose size is a lower bound depending on the elimination order. Computing β_L exactly is MAX-XOR-SAT, NP-hard in general [7]. Our data (§7) suggest β_L as the parameter governing the defect $\min_\sigma \rho(A_\sigma) - 2\sqrt{d-1}$ on cycle-dense instances.

5 The hypercube: an exact certificate

Proposition 5. *On Q_n the system (1) with $L = 4$ is consistent, and every solution σ satisfies $A_\sigma^2 = nI$; consequently $\rho(A_\sigma) = \sqrt{n}$ for the entire affine solution family.*

Proof. Consistency is witnessed by Huang’s signing [9]. Let σ be any solution. Q_n is triangle-free, so $(A_\sigma^2)_{uv} = 0$ for adjacent u, v . If $u \neq v$ are at distance 2 they have exactly two common neighbours w, w' , and $uwvw'$ is a 4-cycle; unbalancedness gives $\sigma_{uw}\sigma_{wv} = -\sigma_{uw'}\sigma_{w'v}$, so the two terms of $(A_\sigma^2)_{uv}$ cancel. Pairs at distance ≥ 3 contribute nothing, and the diagonal equals the degree n . $\square$

This proves the zero variance of ρ observed across sampled solutions on Q_5, Q_6 (all at $\sqrt{5}, \sqrt{6}$ exactly, against a Kesten floor of 4 and $2\sqrt{5}$): on Q_n the local $\mathbb{F}_2$ condition is an exact spectral certificate, far below $2\sqrt{d-1}$. It also isolates what is special: the certificate needs *every* 2-path to be completed by exactly one 4-cycle. Graphs with this property are covered by the two-eigenvalue framework of signed covers; the general question is how much of the cancellation survives when only a β_L -fraction of quadrilaterals can be unbalanced.

6 Failure of two-sided interlacing for A_σ^2

By Godsil-Gutman [6], $\mathbb{E}_\sigma \det(xI - A_\sigma) = \mu_G(x)$, the matching polynomial, which is real-rooted with roots in $[-2\sqrt{d-1}, 2\sqrt{d-1}]$ [8]; this underlies [11]. A natural two-sided analogue would take $q_G(x) := \mathbb{E}_\sigma \det(xI - A_\sigma^2)$ and hope for real-rootedness plus an interlacing family, which would give some σ with $\rho(A_\sigma)^2 \leq \maxroot(q_G)$.

Remark 6. q_G is not real-rooted in general. For $G = C_4$ the two switching classes have A_σ^2 -spectra $\{4, 4, 0, 0\}$ and $\{2, 2, 2, 2\}$, so

$$q_{C_4}(x) = \frac{1}{2}[x^2(x-4)^2 + (x-2)^4] = x^4 - 8x^3 + 20x^2 - 16x + 8 = (x^2 - 4x + 2)^2 + 4,$$

which has no real roots at all. Exhaustive computation over all connected regular graphs on at most 7 vertices, plus Petersen, Q_3 and $K_{3,3}$, shows real-rootedness fails for every instance except the odd cycles (where A_σ^2 is signing-independent) and, curiously, K_4 : $q_{K_4} = x^4 - 12x^3 + 42x^2 - 52x + 21$ is real-rooted despite its three quadrilaterals, so containing an even cycle is not the criterion, and characterizing the graphs with real-rooted q_G is open. In all computed cases every complex root satisfied $\text{Re} \leq 4(d-1)$.

The even-subgraph corrections that cancel in $\mathbb{E} \det(xI - A_\sigma)$ survive squaring; this is the concrete obstruction to extending the interlacing method two-sidedly over the plain signing family, and delimits where any improvement of [13] must diverge from it.

7 Computational evidence

We summarize the computations (code in `signed.spectra.py`; switching classes enumerated exactly as co-tree sign patterns).

Defect atlas. For every connected regular graph on at most 7 vertices and for Petersen, Q_3 , Q_4 , Heawood, Möbius-Kantor and circulants $C_n(1, 2)$, the exact minimum of ρ over all switching classes is at most $2\sqrt{d-1}$, usually strictly below (K_6 : $\sqrt{5} \approx 2.236$ against 3.464, attained by a conference signing; octahedron: 2.0). Among the non-cycle instances, 75-97% of switching classes lie strictly below the floor; for the 2-regular cycles the fraction is 0 or $\frac{1}{2}$, the floor being attained rather than beaten. The exhaustive optimum on Q_4 is Huang's signing. Across all optima, quadrilaterals are unbalanced at rate ≈ 1 , triangles at rate $\approx \frac{1}{2}$, consistent with Corollary 3.

The even-cycle rule at scale. Solving (1) and sampling the affine family: on Q_5, Q_6 all constraints are satisfiable and $\rho = \sqrt{n}$ exactly (Proposition 5); on $C_n(1, 2)$ with n even the quadrilateral system is fully satisfiable and the family sits at $\rho \leq 2\sqrt{2} \approx 2.83$ against a floor of 3.46, an exact evaluation deferred to the companion note [4], and undercutting fairly seeded local search (2.790 against 2.856 for best-of-60 random draws plus local search, on $C_{30}(1, 2)$) at negligible cost. Forcing triangles unbalanced instead *raises* ρ to 3.95, above the random-signing mean 3.44. On a locally tree-like control (random cubic, $n = 60$) all methods land within 0.2 of the Kesten floor and none separates. On K_{12} the system is inconsistent outright (Lemma 4); the maximal simultaneously-unbalanceable fraction satisfies $\beta_4 \geq \frac{1}{2}$ for every graph (average over uniform signings) and $\beta_4 \geq 0.545$ here by local search, and the rule's collapse on dense cliques places them in Seidel-matrix territory.

Variance onset. On symmetry-free instances (random 2-lift towers of Q_3 and K_4 , planted-quadrilateral random 4-regular graphs) we sampled the solution family of (1) alongside uniformly

random signings. The deterministic particular solution produced by row reduction is a strong upward outlier within the family (12-34 standard deviations above the sampled mean on towers, drifting to floor $+0.15$ at $n = 1024$) and must be excluded from dispersion estimates; an earlier draft of this work reported family-to-random dispersion ratios inflated by this outlier. With it excluded, family and random dispersions are comparable on towers (ratios 0.79-1.21 across $n \in \{256, 512, 1024\}$ and all seeds tested): on these dilute instances the constraint system pins a vanishing fraction of the cycle space (8 constraints against 1537 dimensions at $n = 1024$), and conditioning on the short-even signature leaves essentially the fluctuation of a uniform signing. The family minimum over 200 sampled solutions still tracks the Alon-Boppana-forced profile (floor -0.02 at $n = 1024$). The family's advantage over uniform signings is a cycle-density phenomenon: on planted-quadrilateral instances the family minimum beats the random minimum by 0.09-0.23 at equal sample sizes across every size and seed tested, and on circulants random signings concentrate above the Kesten floor from $n \approx 480$ while the family sits at $2\sqrt{2}$ and below. The exact flatness on Q_n is explained by Proposition 5.

Matched-budget baselines, ablation, and generator control. At an equal evaluation budget (2000 spectral-radius evaluations per strategy per instance, 20 instances per set, paired 95% t -intervals, deterministic seeds; `campaign.py`): on planted-quadrilateral 4-regular graphs ($n = 64$) the family's mean gap to the floor is -0.351 ± 0.008 , beating uniform random signings by $+0.094 \pm 0.011$, simulated annealing by $+0.053 \pm 0.010$, and, decisively for attribution, an equal number of uniformly *random* affine $\mathbb{F}_2$ constraints of the same codimension by $+0.098 \pm 0.010$: the effect is parity, not conditioning. (An earlier wall-clock budgeting favoured the search heuristics through the family's per-draw solve overhead; the evaluation budget removes both that bias and the hardware dependence.) A conditioned-uniform control, symmetric double-swap MCMC restricted to the same quadrilateral-count band, reproduces the planted results (-0.340 ± 0.011 , paired deltas agreeing within intervals), excluding generator bias. Tabu search is the one competitive baseline, statistically indistinguishable from the family on the generic sets (-0.003 ± 0.013 , $+0.000 \pm 0.018$): notable, since the family reaches parity with a tuned search while performing no search at all, and family-seeded local search beats fairly seeded random local search at equal draws, as measured above. On circulants the family dominates every arm outright ($+0.17$ to $+0.63$), consistent with its containing the conjecturally optimal class of the companion note.

8 The kernel-averaged trace

Fix a consistent system (1), let $W = \text{span}\{\chi_C\} \subseteq Z_{\text{ev}}$ be the span of its constraint cycles, and let $\mathcal{F}$ be its family of switching classes.

Proposition 7 (Master identity). *The map $\pi: W \rightarrow \mathbb{F}_2$, $\pi(\sum_{C \in S} \chi_C) = |S| \bmod 2$, is a well-defined linear form, and for every even ℓ ,*

$$\mathbb{E}_{\sigma \in \mathcal{F}} \text{tr}(A_{\sigma}^{\ell}) = \sum_{z \in W} (-1)^{\pi(z)} N_{\ell}(z).$$

Proof. Consistency means every dependency $\sum_{C \in S} \chi_C = 0$ has $|S|$ even, so π is well defined; linearity follows from $|S| + |S'| \equiv |S \Delta S'| \pmod{2}$. Identify switching classes with $\text{Hom}(Z(G), \mathbb{F}_2)$; then $\mathcal{F} = \varphi_0 + \text{Ann}(W)$ and $\varphi|_W = \pi$ for every $\varphi \in \mathcal{F}$. For $z \in Z(G)$ the average $\mathbb{E}_{\psi \in \text{Ann}(W)} (-1)^{\psi(z)}$ equals 1 if $z \in (\text{Ann } W)^{\perp} = W$ and 0 otherwise. Averaging the

expansion of Lemma 1 term by term gives the claim; note $W \subseteq Z_{\text{ev}}$ since the constraint cycles are even. $\square$

Walks whose wrap parity touches any cycle outside W are annihilated; a single constraint cycle enters with -1 , a pair with $+1$, and so on.

Corollary 8. *For every k ,*

$$\min_{\sigma \in \mathcal{F}} \rho(A_\sigma) \leq R_{\mathcal{F}}(k) := \left(\sum_{z \in W} (-1)^{\pi(z)} N_{2k}(z) \right)^{1/2k}.$$

By contrast the average over all signings retains only $z = 0$: $\mathbb{E}_\sigma \text{tr}(A_\sigma^{2k}) = N_{2k}(0) \geq n t_{2k}(d)$, where $t_{2k}(d)$ counts closed walks at the root of the infinite d -regular tree; indeed every such tree walk crosses each tree edge an even number of times and projects, injectively by unique lifting, to a closed walk in G with all edge multiplicities even. Hence the uniform average can never certify a spectral radius below the Kesten profile, while $R_{\mathcal{F}}(k)$ can, if and only if the signed sum over $W \setminus \{0\}$ is negative enough to consume both the even-wrap excess $N_{2k}(0) - n t_{2k}(d)$ and part of the tree term.

Corollary 9 (Confined mass). *The homogeneous family $\text{Ann}(W)$, all constraint cycles balanced, measures the total parity-confined walk mass:*

$$\mathbb{E}_{\psi \in \text{Ann}(W)} \text{tr}(A_\psi^{2k}) = \sum_{z \in W} N_{2k}(z).$$

This is the same annihilator computation with $\varphi_0 = 0$, so every $z \in W$ enters with sign $+1$; it makes the confined mass, and hence Question 11 below, directly measurable by sampling eigenvalues.

Numerically the mechanism is already decisive at $k = 40$ with 400 sampled family members ($n = 128$). On a random 2-lift tower of K_4 ($d = 3$, Kesten floor $2\sqrt{2} \approx 2.828$): $R_{\mathcal{F}}(40) \approx 2.769$ against $R_{\text{all}}(40) \approx 2.846$ and tree rate 2.839; the kernel-averaged trace alone certifies a sub-Kesten signing on a symmetry-free instance, which Corollary 8 shows the uniform average can never do. On a planted-quadrilateral 4-regular graph (37 of 41 quadrilateral constraints in a greedy consistent subsystem, floor 3.464): $R_{\mathcal{F}}(40) \approx 3.327$. On $C_{60}(1, 2)$ the family average is dominated by its $2\sqrt{2}$ -class, $R_{\mathcal{F}}(40) \approx 2.872$, while R_{all} stalls near 3.68, above the tree rate, as the theory demands.

Measuring all three averages against the tree profile $n t_{2k}(d)$ separates the confined masses. On random 2-lift towers the number of short cycles stays bounded as n grows, so the confined mass dilutes and the instances drift toward the bicycle-free regime; the demanding test is constant local density. On planted-quadrilateral 4-regular graphs at fixed density $\frac{1}{2}$ quadrilateral per vertex, $n \in \{128, 256, 512\}$, $k = 30$: the ordering $R_{\mathcal{F}} < R_{\text{all}} \approx R_{\text{tree}} < R_{\text{bal}}$ holds throughout; the confined-rate excess $R_{\text{bal}} - R_{\text{all}} \approx 0.12\text{--}0.15$ is n -independent; the confined-mass ratio $\mathbb{E}_{\text{bal}}/\mathbb{E}_{\text{all}}$ remains $O(1)$ in n at fixed k ; and $\mathbb{E}_{\mathcal{F}} \text{tr}(A^{2k})/(n t_{2k}) \approx 0.08\text{--}0.11$, below one with an n -stable margin: the parity-corrected average beats the tree profile itself. These are precisely the behaviours the counting question below predicts, with rate excess $\delta \approx 0.04$ at this density.

9 A kernel-averaged L -function

By Bass's theorem [2], for any signing $\det(I - uB_\sigma) = \prod_p (1 - \sigma(p)u^{|p|})$, the product running over primes (primitive cyclic classes of closed non-backtracking walks), where $\sigma(p) = (-1)^{\varphi(z(p))}$ and $z(p) \in Z(G)$ is the edge-multiplicity parity vector of p . Averaging over the family diagonalizes this product.

Proposition 10 (Kernel-averaged L -identity). *With W , π , $\mathcal{F}$ as above, $\varepsilon_p = (-1)^{\pi(z(p))}$, and B the unsigned non-backtracking operator, for $|u| < \frac{1}{d-1}$,*

$$\mathbb{E}_{\sigma \in \mathcal{F}} \log \det(I - uB_\sigma) = \frac{1}{2} \log \det(I - u^2B) - \frac{1}{2} \sum_{p: z(p) \in W} \log \frac{1 + \varepsilon_p u^{|p|}}{1 - \varepsilon_p u^{|p|}}.$$

Proof. As in Proposition 7, $\varphi(z)$ is frozen at $\pi(z)$ over $\mathcal{F}$ when $z \in W$ and is a fair coin otherwise. Hence $\mathbb{E} \log(1 - \sigma(p)u^{|p|})$ equals $\log(1 - \varepsilon_p u^{|p|})$ for parity-confined primes and $\frac{1}{2} \log(1 - u^{2|p|})$ for the rest. Collecting the coin factor over *all* primes gives $\frac{1}{2} \log \prod_p (1 - (u^2)^{|p|}) = \frac{1}{2} \log \det(I - u^2B)$, and the per-prime correction for confined primes is $\log(1 - \varepsilon u^\ell) - \frac{1}{2} \log(1 - u^{2\ell}) = \frac{1}{2} \log \frac{1 - \varepsilon u^\ell}{1 + \varepsilon u^\ell}$. Absolute convergence for $|u|(d-1) < 1$ follows from $\#\{p : |p| = \ell\} \leq \frac{2m}{\ell}(d-1)^\ell$. $\square$

Both sides transfer to the adjacency side member-wise via $\det(I - uB_\sigma) = (1 - u^2)^{m-n} \det(I - uA_\sigma + u^2(d-1)I)$. The identity was verified exactly on the prism ($\mathcal{F}$ of size 2, W spanned by the three quadrilaterals): the coefficient form $\mathbb{E}_{\mathcal{F}} \text{tr } B_\sigma^k$ matches the prime decomposition for all $k \leq 14$ to machine precision, and the displayed identity holds to 8 digits at $u = 0.2$ under truncation at $|p| \leq 14$.

The structural content: $\frac{1}{2} \log \det(I - u^2B)$ is analytic exactly in $|u| < (d-1)^{-1/2}$, the Ramanujan radius; *primes whose parity escapes W contribute the Ramanujan rate automatically, and the entire deviation of the averaged L -function from that rate is carried by parity-confined primes.* At the level of the averaged L -function, the sign problem has become a counting problem. We emphasize the scope of the identity: the domain of analyticity of $\mathbb{E} \log \det$ is governed by the worst member of $\mathcal{F}$, so it does not by itself bound $\min_{\sigma \in \mathcal{F}} \rho$; the quantitative route of this paper is Corollary 8, and the identity's role is to explain the mechanism and to isolate Question 11.

Question 11 (Confined-prime counting). For which graph sequences does $\#\{p : |p| = \ell, z(p) \in W\} \leq C(\sqrt{d-1} + \delta)^\ell$ hold? Parity confinement means the odd-multiplicity edge set of p is a sum of short even cycles; the extreme case $z(p) = 0$ (totally even primes) counts non-backtracking closed walks retracing every edge evenly, which grow at rate $\sqrt{d-1}$, and each nonzero $z \in W$ should contribute a comparable rate with an exponential penalty in $|z|$. Caution from the prism: on graphs so small that W nearly fills the cycle space, confinement is no restriction (measured rate ≈ 1.84 against $\sqrt{2} \approx 1.41$ at $d = 3$); the question is asymptotic, with $\dim W$ per ball bounded while $\dim Z(G) \rightarrow \infty$.

10 The $z = 0$ slice: totally even walks

Sub-question of Question 11 at $z = 0$: count closed non-backtracking walks all of whose edge multiplicities are even ("totally even" walks). The lower bound holds at the conjectured rate unconditionally; the upper bound fails without a hypothesis on dense cores, and the failure mechanism is instructive.

Proposition 12 (Slice identity and doubling). *Let Ev_ℓ denote the number of totally even closed non-backtracking walks of length ℓ in G . Then $\text{Ev}_\ell = \mathbb{E}_\sigma \text{tr}(B_\sigma^\ell)$, the average over all signings; and the doubling map $w \mapsto w^2$ injects closed non-backtracking walks of length $\ell/2$ into totally even walks of length ℓ , so $\text{Ev}_\ell \geq \text{tr}(B^{\ell/2})$ and $\liminf_\ell \text{Ev}_\ell^{1/\ell} \geq \rho(B)^{1/2} = \sqrt{d-1}$.*

Proof. $\text{tr}(B_\sigma^\ell)$ is the signed count of closed non-backtracking walks, the sign being $\prod_e \sigma_e^{m_e(w)}$; averaging kills every walk with an odd multiplicity. If w is cyclically non-backtracking then so is w^2 (the seam repeats a legal transition of w), its multiplicities double, and w is recovered as the first half. $\square$

Proposition 13 (Dense trapping). *Let $d \geq 4$ and suppose G contains K_d as a subgraph (realizable in connected d -regular graphs: two copies of K_d joined by a perfect matching). Then there are $c_d > 0$ and $\ell_0(d)$, independent of G and n , with $\text{Ev}_\ell(G) \geq c_d(d-2)^\ell$ for all even $\ell \geq \ell_0$. Since $d-2 > \sqrt{d-1}$ for every $d \geq 4$, the confined mass exceeds the profile $n(\sqrt{d-1})^\ell$ throughout $\ell \geq C_d \log n$.*

Proof. Totally even walks of the K_d copy are totally even walks of G , and by Proposition 12 their number is $2^{-|E(K_d)|} \sum_{\sigma \in \{\pm 1\}^{E(K_d)}} \text{tr}(B_{K_d, \sigma}^\ell)$. The unsigned B_{K_d} is irreducible and aperiodic with Perron value $d-2$; its remaining eigenvalues have modulus at most $\max(1, \sqrt{d-2}) < d-2$ (Ihara-Bass from $\text{spec } A(K_d) = \{d-1, (-1)^{d-1}\}$). By the equality case of Wielandt's theorem, a signed version satisfies $\rho(B_{K_d, \sigma}) = \rho(B_{K_d})$ if and only if $B_{K_d, \sigma} = \pm D B_{K_d} D^{-1}$ for a diagonal D with $|D| = I$, i.e. iff σ is switching-equivalent to the all-plus or all-minus signing. Hence $\rho_2 := \max$ over the remaining classes is strictly below $d-2$, the two trivial classes contribute $\text{tr } B^\ell + (-1)^\ell \text{tr } B^\ell = 2(d-2)^\ell(1 + o(1))$ at even ℓ , and the class average is at least $2^{-r} [2(d-2)^\ell(1 - o(1)) - 2^r d(d-1)\rho_2^\ell]$ with $r = \binom{d}{2} - d + 1$. $\square$

Remark 14 (Necessity of a core hypothesis). Since $0 \in W$ always, Proposition 13 shows the bound of Question 11 cannot hold unconditionally once $\ell \geq C_d \log n$: it forces the exclusion of K_d ; for $d=4$ the same K_4 as Lemma 4, which thus obstructs the program twice. Any workable hypothesis must constrain the *unsigned* local growth, that is, the closed-walk counts $c_H(t)$ of condition (ii) in the subcriticality definition of Section 10: the trapped mass of Proposition 13 is carried by the two $\pm$ -trivial classes with only the constant discount $2^{-\binom{d}{2} + d - 1}$, so a hypothesis exempting the trivial classes would be vacuous, since every non- $\pm$ -trivial class of K_4 and of K_5 has $\rho(B_\sigma)$ equal to $\sqrt{d_H - 1}$ exactly ($\sqrt{2}$ and $\sqrt{3}$), so K_4 would satisfy such a hypothesis at $d=4$ while violating the conclusion. K_d does violate condition (ii), since $c_{K_d}(t)$ grows at rate $d-2 > \sqrt{d-1}$. On the Petersen graph every non-trivial class likewise sits at $\rho(B_\sigma) = \sqrt{2}$: every nontrivial signing of K_4 and of Petersen is exactly Ramanujan.

The upper bound matching Proposition 12 holds in the dilute regime. Say G is (r_0, δ) -subcritical at scale ℓ if every connected subgraph H of G with minimum degree ≥ 2 and at most ℓ edges satisfies: (i) the cycle rank of H is at most r_0 ; and (ii) the number $c_H(t)$ of closed non-backtracking walks of length t in H obeys $c_H(t) \leq 2|E(H)|(t+1)^{3r_0}(1+\delta)^t$ for all $t \leq 8\ell$. The notion is monotone: subcriticality at scale ℓ implies it at every smaller scale, and all subsequent hypotheses are stated through this single definition. Condition (ii) holds, for instance, whenever every thread of H (a maximal path through degree-2 vertices) has length at least λ , with $1 + \delta = (2r_0 - 1)^{1/\lambda}$: closed walks in H project to reduced cyclic words in a free group of rank $\leq r_0$ whose generators cost at least λ steps each.

Lemma 15 (Non-backtracking reachability). *Let H be connected with minimum degree ≥ 2 , s edges, and not a cycle. Between any two directed edges of H there is a non-backtracking path of length at most $4s$.*

Proof. The non-backtracking digraph of a connected graph with minimum degree ≥ 2 that is not a cycle is strongly connected [10]; concretely, a theta or dumbbell subgraph, present because some vertex has degree ≥ 3 , lets a walk reverse its direction of travel along any thread. A strongly connected digraph on the $2s$ directed edges has diameter at most $2s - 1 \leq 4s$. $\square$

Proposition 16 (Even-walk upper bound, dilute regime). *If G is d -regular and (r_0, δ) -subcritical at scale ℓ , then*

$$\text{Ev}_\ell(G) \leq nd (C\ell)^{3(r_0+1)^2} \left((1 + \delta)^{2(r_0+2)} \sqrt{d-1} \right)^\ell$$

for an absolute constant C . Combined with Proposition 12, in this regime $\text{Ev}_\ell = (\sqrt{d-1})^{\ell(1+o(1))}$: the $z = 0$ case of Question 11 is settled.

Proof. Fix the starting directed edge ($\leq nd$ choices) and expose the walk step by step; let S_i be the subgraph of edges used through step i and $S = S_\ell$ the support. Since multiplicities are even, $s := |E(S)| \leq \ell/2$; since a degree-1 vertex of S would force a backtrack, S has minimum degree 2; by (i) its cycle rank r is at most r_0 .

Call a step *fresh* if it traverses an edge not in S_{i-1} , *stale* otherwise. The key structural fact: after a fresh step into a previously unvisited vertex v , the only used edge at v is the arrival edge, which is excluded by non-backtracking, so every continuation is again fresh. Hence maximal fresh runs continue until they land on an already-visited vertex, i.e. each maximal fresh run is an open ear attached to the current support, and the number of fresh runs equals the number of ears, which is exactly the cycle rank: *the walk performs precisely $r \leq r_0$ fresh runs* (the first builds the initial cycle), interleaved with at most $r + 1$ maximal stale segments.

Encode the walk by: the start edge; the set of run-start times, at most $\binom{\ell}{r} \leq \ell^{r_0}$ choices (run *ends* are visible to a decoder who knows G and the visited set: a run ends exactly when its landing vertex is old); the fresh-edge choices, at most $(d-1)^s$ in total; and the stale segments. A stale segment of length t is a non-backtracking path inside S , and by Lemma 15 any such path prolongs to a closed walk of length $\leq t + 4s$, so their number is at most $2s \cdot c_S(t + 4s) \leq 4s^2(t + 4s + 1)^{3r_0} (1 + \delta)^{t+4s}$ by (ii) (if S is a cycle the count is 1). Multiplying over the $\leq r_0 + 1$ segments with total stale length $\ell - s$ and collecting,

$$\text{Ev}_\ell \leq nd \ell^{r_0} \sum_{s \leq \ell/2} (4s^2 (4\ell)^{3r_0})^{r_0+1} (d-1)^s (1 + \delta)^{(\ell-s)+4s(r_0+1)}.$$

Every factor of the summand is nondecreasing in s , so it is maximized at $s = \ell/2$, where $(d-1)^{\ell/2} (1 + \delta)^{\ell/2+2\ell(r_0+1)} \leq ((1 + \delta)^{2(r_0+2)} \sqrt{d-1})^\ell$; the polynomial prefactors collect into $(C\ell)^{3(r_0+1)^2}$. $\square$

Remark 17 (Scope). Random 2-lift towers are $(O(1), o(1))$ -subcritical at scale $c \log n$ with high probability: balls of that radius have bounded cycle rank and long threads, and the measured $\mathbb{E}_\sigma[\text{tr } B_\sigma^\ell]^{1/\ell} \approx 1.53$ at $\ell = 30$, $n = 128$, $d = 3$ against $\sqrt{2}$ is consistent with the proposition's $(1 + \delta)$ -corrected rate. Constant cycle density (quadrilaterals proportional to n) violates hypothesis (i): supports with s edges can have rank $\Theta(\eta s)$. The run-schedule cost survives this: $\binom{\ell}{r} \leq (e\ell/r)^r = (1 + o_\eta(1))^\ell$ for $r \leq \eta\ell$, so the sole obstruction to the constant-density extension

is the stale-path entropy bound for unbounded-rank supports, which is an unsigned growth requirement in the spirit of condition (ii); Remark 14 explains why any such bound must constrain the trivial classes. That extension is carried out below via Lemmas 19 and 20.

Remark 18 (Toward constant density: the critical gadget). At $d = 3$ the candidate obstruction to the unbounded-rank entropy bound is a quadrilateral with exits at two opposite vertices: oscillation between its branch points along the two length-2 arcs would give path growth $2^{2/4} = \sqrt{2} = \sqrt{d-1}$, exactly critical. Non-backtracking forbids it: arriving along one short arc, a walk may not reverse into it, so bouncing between the two short arcs is deterministic, and branching entropy is carried only by the long arm. Exactly: on the theta graph $\Theta(2, 2, \lambda)$ every switching class satisfies $\rho(B_\sigma) \leq \rho(B) = 1 + O(1/\lambda)$ (measured 1.348, 1.268, 1.222, 1.190, 1.150 for $\lambda = 3, 5, 7, 9, 13$, all nontrivial classes strictly below), and a ring of four such quadrilaterals separated by λ -threads behaves identically ($\rho \leq 1.25$ at $\lambda = 6$, rank 5). The general principle: if every ball of radius R contains at most one cycle, then at any branch vertex of any support at most one continuation returns within $2R$ steps; two short returns would exhibit two cycles in one ball. The remaining entropy bound therefore takes the quantified form: paths in any support grow at rate $1 + O(\log d/R)$; ranks obey $r \leq C\ell/R$, keeping the run-schedule cost $\binom{\ell}{r}$ at $(1 + o(1))^\ell$; and the per-segment polynomial factors force $R \geq C \log \log n/\delta$, the scale at which the bicycle-free hypothesis of [12] reappears, now in service of the parity family rather than the uniform measure.

The entropy half of that program is now a lemma.

Lemma 19 (Window path bound). *Suppose every ball of radius R in G contains at most one cycle. Then for every subgraph $S \subseteq G$ with s edges, every directed edge of S , and every $t \geq 1$, the number of non-backtracking paths of length t in S from that edge is at most $(2s(R+3))^{\lceil t/R \rceil}$; in particular the growth rate is $\exp(O(\log(sR)/R))$.*

Proof. Cut the path into $\lceil t/R \rceil$ windows of at most R consecutive steps. A window starting at the vertex w stays inside $B_R(w)$, so its segment is a non-backtracking path in $H_w := S \cap B_R(w)$, a graph of cycle rank at most 1. In a tree a non-backtracking walk never revisits a vertex: a revisit would have to undo an edge crossing, and the first undoing step is a backtrack, so it is the unique reduced path between its endpoints. In a unicyclic graph with cycle C , each component of the complement of $E(C)$ is a tree attached to C at a single vertex; once a non-backtracking walk enters such a tree it can never return to the attachment vertex, and while on C its rotational direction cannot reverse. A non-backtracking path therefore consists of a unique reduced tree path to an attachment vertex, unidirectional motion around C with some winding number $w \leq t/3$, and a unique reduced tree path to its final edge: it is determined by its endpoint directed edges, the direction, and w , giving at most $2(1 + t/3)$ paths per endpoint pair (the entrywise count is at most $1 + \lceil t/\text{girth} \rceil$, numerically sharp). Each window therefore admits at most $4s(R/3 + 2) \leq 2s(R+3)$ continuations given its starting edge, using $R \geq 3$. $\square$

The rate is sharp in the exponent. The *tree-burst* gadget, two depth- D binary trees with roots and matched leaves joined by λ -threads, every cycle of length $\geq 2\lambda + 4$, hence bicycle-free at radius λ , has measured $\rho(B)$ tracking the burst rate $2^{D/(D+\lambda)}$: (1.29, 1.19, 1.34, 1.22, 1.25) against (1.35, 1.21, 1.41, 1.26, 1.31) for $(D, \lambda) = (3, 4), (3, 8), (4, 4), (4, 8), (5, 8)$. Since D can reach $\log_2(s/\lambda)$, path growth $\exp(c \log s/R)$ is realized, and $R \geq C \log \ell/\delta$ is *necessary* for a $(1 + \delta)^t$ bound, matching Lemma 19.

Lemma 20 (Rank of bicycle-free graphs). *There is an absolute constant C such that every graph H with $s \geq R \geq 8$ edges and minimum degree ≥ 2 that is bicycle-free at radius R satisfies $\text{rank}(H) \leq 1 + Cs \log s/R$.*

Proof. Call a cycle *short* if its length is at most $2R$. Two short cycles cannot share a vertex, since both would lie in that vertex's R -ball; hence short cycles are pairwise vertex- and edge-disjoint, and there are at most s/R of them with length in $(R, 2R]$. Two disjoint short cycles of length $\leq R$ at distance ρ have $\rho > R/2$: otherwise, taking v on one nearest the other, every vertex of both lies within $\rho + R/2 \leq R$ of v , giving a rank-2 ball. Their $R/4$ -balls are then disjoint, and each contains at least $R/4$ edges by minimum degree 2, so there are at most $4s/R$ of them. Delete one edge from each short cycle: $D \leq 5s/R$ deletions destroy every short cycle and create none, so the result H' has girth exceeding $2R$. Pass to the 2-core of each component of H' ; this changes neither rank nor girth. By the Moore bound for irregular graphs [1], a graph with minimum degree ≥ 2 , girth $\geq 2R+1$, and average degree $\bar{d}_i$ on n_i vertices satisfies $n_i \geq (\bar{d}_i - 1)^R$ for either girth parity, so $\bar{d}_i - 1 \leq n_i^{1/R}$, whence $\text{rank}(H') \leq \sum_i \left[\frac{n_i}{2} (n_i^{1/R} - 1) + 1 \right]$. If $\log s \leq R$ this is at most $Cs \log s/R + D + 1$, using $e^x - 1 \leq 2x$ on $[0, 1]$, $\sum_i n_i \leq s$, and $\#\text{components} \leq D + 1$; if $\log s > R$ the claimed bound exceeds s and is trivial. Finally $\text{rank}(H) \leq \text{rank}(H') + D$. $\square$

The $\log s$ factor is necessary, and the earlier guess $\text{rank} \leq 1 + Cs/R$ is false for every constant C : a graph of girth $\geq 2R + 2$ has every R -ball a tree, hence is bicycle-free at radius R , while its rank density is $1 - 2/\bar{d}$. The Tutte-Coxeter $(3, 8)$ -cage is bicycle-free at radius 3 with $(\text{rank} - 1)/s = \frac{1}{3}$, the girth-12 Tutte cage is bicycle-free at radius 5 at the same $\frac{1}{3}$, and generalized-polygon cages of degree $\bar{d} \sim s^{1/(R+1)}$ push the density toward 1 (such cages exist for $R \in \{2, 3, 5\}$ by Feit-Higman; for other R , girth- $(2R + 2)$ graphs of growing degree serve the same purpose). (Stochastic search over bicycle-free graphs found exactly this ceiling, 0.344 at $R = 3$ in small sizes, after first refuting the linear conjecture's banana extremal at density $1/(R + 1)$.) At the operating scale the lemma gives precisely what is needed: for $s \leq \ell$ and $R \geq C \log \ell/\delta$, $\text{rank} \leq 1 + \delta \ell$.

Proposition 21 (ε -theorem, bicycle-free regime). *Fix $\delta \in (0, 1)$, L , Λ , and let G be d -regular on n vertices, bicycle-free at radius $R \geq C \log \log n/\delta$, with W spanned by any consistent family of even cycles of length $\leq L$ and $\Lambda_W \leq \Lambda$. Then some signing in the solution family satisfies $\rho(A_\sigma) \leq 2\sqrt{d-1} (1 + C\delta \log(1/\delta))(1 + o(1))$.*

Proof. Set $\ell = 2\lceil \log n \rceil$ and $k = \lceil \log n \rceil$; we may assume $R \leq \ell$, since otherwise every support below is bicycle-free at radius exceeding its size and Proposition 24 applies with $r_0 = 1$. Fix $j \leq 2\ell$ and bound $\sum_{z \in W} N_j^{\text{nb}}(z)$ by rerunning Propositions 16 and 22. A counted walk has support S with $s \leq (j + \Lambda)/2 \leq 2\ell$ edges and minimum degree 2; by Lemma 20, its cycle rank satisfies $r \leq 1 + Cs \log(2\ell)/R \leq 1 + \delta(j + \Lambda)$ once $R \geq C' \log \log n/\delta$. The fresh-run identity is untouched: the walk performs exactly r fresh runs, so the run-start schedule costs $\binom{j}{r} \leq (ej/r)^r \leq (e/\delta)^{\delta(j+\Lambda)+1}$, and the fresh choices cost $(d-1)^s \leq (d-1)^{\Lambda/2} (d-1)^{j/2}$. Each of the at most $r + 1$ stale segments is a non-backtracking path in S , and Lemma 19 bounds its continuations by $(2s(R + 3))^{\lceil t_i/R \rceil}$ with $\sum_i t_i \leq j$, for a total window exponent at most $j/R + \delta(j + \Lambda) + 3$. Hence

$$\sum_{z \in W} N_j^{\text{nb}}(z) \leq K \hat{\rho}^j, \quad \hat{\rho} := (1 + \eta)\sqrt{d-1},$$

where $K \leq nd(d-1)^{\Lambda/2}(e/\delta)^{\delta\Lambda+1}(4\ell(R+3))^{\delta\Lambda+3}$ collects all factors not exponential in j , and

$$\log(1+\eta) = \delta \log(e/\delta) + \left(\frac{1}{R} + \delta\right) \log(4\ell(R+3)) \leq C\delta \log(1/\delta)$$

at $R \geq C' \log \log n / \delta$ and $R \leq \ell$. The peeling and transfer of Lemma 23 and the assembly in the proof of Proposition 24 now apply verbatim with this K and $\hat{\rho}$, giving $\mathbb{E}_{\mathcal{F}} \text{tr } A_{\sigma}^{2k} \leq C'' K (Ck)^2 ((1+\eta)^2 2\sqrt{d-1})^{2k}$, and Corollary 8 at $k = \lceil \log n \rceil$ yields $\min_{\sigma \in \mathcal{F}} \rho(A_{\sigma}) \leq 2\sqrt{d-1} (1+\eta)^2 (C'' K (Ck)^2)^{1/2k}$, which is the claim. $\square$

The extension from $z = 0$ to all of W costs only a constant. Set $\Lambda_W := \max_{z \in W} |z| \leq L \dim W$.

Proposition 22 (Confined-walk upper bound). *Assume $\Lambda_W \leq \ell$ and that G is (r_0, δ) -subcritical at scale ℓ . Then, with $N_{\ell}^{\text{nb}}(z)$ the number of closed non-backtracking walks of length ℓ and parity z ,*

$$\sum_{z \in W} N_{\ell}^{\text{nb}}(z) \leq C_{r_0, \delta, \Lambda_W, d} n (C\ell)^{3(r_0+1)^2} ((1+\delta)^{2(r_0+2)} \sqrt{d-1})^{\ell}.$$

Proof. A counted walk has $|z(w)| \leq \Lambda_W$; edges of $z(w)$ carry odd multiplicity ≥ 1 and all others even multiplicity ≥ 2 , so its support has $s \leq (\ell + \Lambda_W)/2$ edges and, as before, minimum degree 2. The proof of Proposition 16 used evenness only through the bound on s ; it applies verbatim, with $(d-1)^s \leq (d-1)^{\Lambda_W/2} (d-1)^{\ell/2}$ and the corresponding constant absorbed into $C_{r_0, \delta, \Lambda_W, d}$. $\square$

To reach the adjacency side, we use the classical linearization (see e.g. [5]). Define $A_0 = I$, $A_1 = A_{\sigma}$, $A_2 = A_{\sigma}^2 - dI$, and $A_{j+1} = A_{\sigma} A_j - (d-1)A_{j-1}$ for $j \geq 2$; then $(A_j)_{uv}$ is the signed count of non-backtracking walks $u \rightarrow v$ of length j , and inverting the recursion, $A_{\sigma}^m = \sum_{j \leq m} \alpha_{m,j} A_j$ where $\alpha_{m,j} \geq 0$ counts weight- $(d-1)$ downward-step walks on $\mathbb{N}$ from 0 to j . Both identities and the envelope below were verified symbolically through $m = 8$.

Lemma 23 (Transfer envelope and peeling). (i) $\sum_j \alpha_{m,j} (\sqrt{d-1})^j \leq \frac{d}{d-1} (m+1) (2\sqrt{d-1})^m$. (ii) Let $\tilde{N}_j(z)$ count vertex-based non-backtracking closed walks (backtracking permitted at the wrap) of length j and parity z . Then $\tilde{N}_j(z) \leq \sum_{p \geq 0} d(d-1)^p N_{j-2p}^{\text{nb}}(z) + d(d-1)^{j/2-1} \mathbf{1}_{z=0}$.

Proof. (i) Each $\mathbb{N}$ -walk of length m ending at j has $(m-j)/2$ down-steps, so $\alpha_{m,j} \leq \binom{m}{(m+j)/2} (d-1)^{(m-j)/2} \cdot \frac{d}{d-1} \leq \frac{d}{d-1} 2^m (d-1)^{(m-j)/2}$; multiply by $(\sqrt{d-1})^j$ and sum. (ii) A vertex-based walk is either cyclically non-backtracking or of the form $e \cdot w' \cdot \bar{e}$ with w' vertex-based of length $j-2$ at the head of e ; peeling the pendant edge adds 2 to the multiplicity of e , leaving the parity unchanged. Iterate: a pendant path of length p costs at most $d(d-1)^{p-1}$ choices, and the residue is cyclically non-backtracking (or empty, forcing $z = 0$). $\square$

Proposition 24 (ε -theorem, dilute regime). *Fix r_0, δ, L, Λ and let G be d -regular on n vertices, (r_0, δ) -subcritical at scale $\ell = 2\lceil \log n \rceil$. Let W be spanned by any consistent family of even cycles of length $\leq L$ with $\Lambda_W \leq \Lambda$, possibly $W = \{0\}$. Then some signing in the solution family $\mathcal{F}$ satisfies*

$$\rho(A_{\sigma}) \leq 2\sqrt{d-1} (1+\delta)^{4(r_0+2)} \left(1 + O_{r_0, \delta, \Lambda, d} \left(\frac{\log \log n}{\log n}\right)\right).$$

Proof. For n large enough that $\Lambda \leq \ell$, Proposition 22 applies at every length $j \leq 2k$ by monotonicity of the subcriticality notion. Write $\hat{\rho} = (1 + \delta)^{2(r_0+2)}\sqrt{d-1}$ and let K denote the prefactor of Proposition 22. Since the recursion defining A_j preserves the signed-walk interpretation, the annihilator computation gives $\mathbb{E}_{\text{Ann}(W)} \text{tr } A_j(A_\sigma) = \sum_{z \in W} \tilde{N}_j(z)$. By Lemma 23(ii), Proposition 22, and $\hat{\rho}^2 \geq d-1$ (terms nonincreasing in p), $\sum_{z \in W} \tilde{N}_j(z) \leq (j+2) dK \hat{\rho}^j$. Hence with $k = \lceil \log n \rceil$,

$$0 \leq \mathbb{E}_{\mathcal{F}} \text{tr } A_\sigma^{2k} \leq \mathbb{E}_{\text{Ann}(W)} \text{tr } A_\sigma^{2k} = \sum_j \alpha_{2k,j} \sum_{z \in W} \tilde{N}_j(z) \leq (2k+2)dK \sum_j \alpha_{2k,j} \hat{\rho}^j,$$

and by Lemma 23(i) applied after extracting $(1 + \delta)^{2(r_0+2)j} \leq (1 + \delta)^{4(r_0+2)k}$, the right side is at most $C' n (Ck)^{3(r_0+1)^2+2} ((1 + \delta)^{4(r_0+2)} 2\sqrt{d-1})^{2k}$. Corollary 8 and $(n(Ck)^c)^{1/2k} = 1 + O(\log \log n / \log n)$ conclude. $\square$

Taking $W = \{0\}$ recovers a near-Ramanujan statement for random signings of dilute graphs in the spirit of [12], from entirely different machinery; taking W maximal uses the full parity family. Random 2-lift towers satisfy the hypotheses with $r_0 = O(1)$, $\delta = o(1)$, $\Lambda = O(1)$ with high probability, so the ε -version of the Bilu-Linial conjecture holds along them.

11 Outlook

The target is now a single inequality: show $R_{\mathcal{F}}(k) \leq 2\sqrt{d-1} + \varepsilon(d, \beta_L)$ at $k \asymp \log n$, i.e. that the single-cycle classes $\sum_C N_{2k}(\chi_C)$ dominate the even-wrap excess $N_{2k}(0) - n t_{2k}(d)$ while the pair classes stay controlled. The structure of Proposition 7 is that of an Ihara-Bass/cluster expansion in which the local factors of the constraint cycles appear with inverted sign; Proposition 10 makes this exact, and isolates the analytic core of the program in Question 11, with the bicycle-free walk taxonomy of [12] supplying the combinatorics and now entering with cancellation rather than by union bound. The variance-onset data locate the regime boundary: on dilute instances the family fluctuates like a uniform signing, and its measurable advantage is confined to cycle-dense instances, precisely where the parity weights of Proposition 7 act.

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